2D-PDM: three-stream RGB-D framework

Single-view scene observation

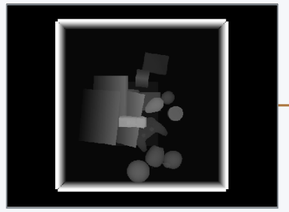
1. Inputs

Scene RGB I



640 × 480

Observed depth D



Registered to RGB

Target reference



RGB thumbnail + text



Reference mask

Text prompt

2a. Similarity stream

Trained

DINOv3 scene + target RGB (F)

Four scene feature layers

target

SigLIP target RGB + text (F)

Masked pooling + semantic adapter (T)

reference mask

Multi-layer matching + feature CNN (T)

F_S

64×30×40

1×1+σ

S map



2b. Occlusion stream

Trained

DINOv3 scene + target RGB (F)

Four scene feature layers

Target: full-frame mean pooling

Depth ResNet18 depth + valid mask (T)

Global FiLM geometry 68 + MLP (T)

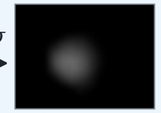
Multi-layer matching + feature CNN (T)

F_O

64×30×40

1×1+σ

O map



workspace mask

2c. Complexity stream

Trained

DINOv3 RGB last-layer features (F)

Layer 11 features

RGB CNN (T)

MultiMAE observed-depth encoder (F)

Depth encoding; no depth prediction

Projection + concat dilated CNN (T)

F_C

64×30×40

Softplus

C map



3. Feature fusion

Proposed; not yet trained

Concatenate stream features

$[F_S; F_O; F_C]$

192 × 30 × 40

CNN fusion head architecture to finalize

Amodal target map

Planned prediction

supervision

Amodal GT



Full projected target mask including occluded parts

Stream training Separate supervision with similarity, occlusion and regional-complexity GT.

F Frozen pretrained backbone **T** Learned stream module

Feature tiles are schematic; map thumbnails are actual test predictions.

Map outputs: 1 × 30 × 40. Shared DINO weights; current streams run independently.

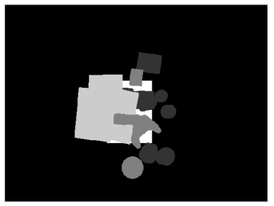
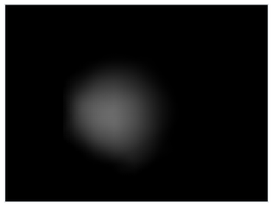
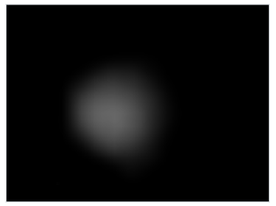
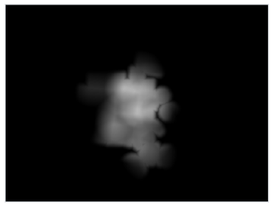
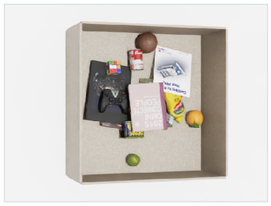
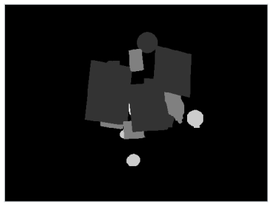
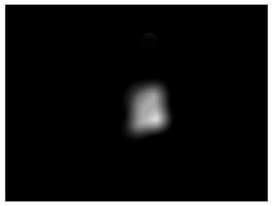
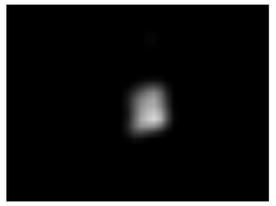
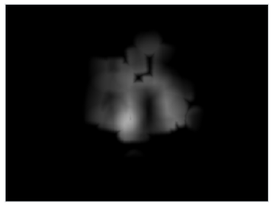
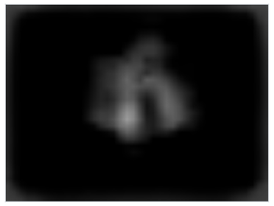
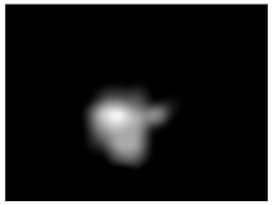
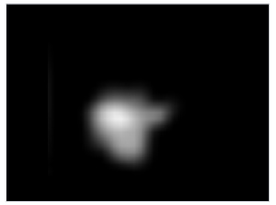
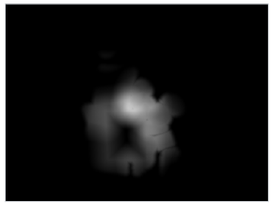
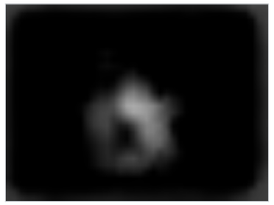
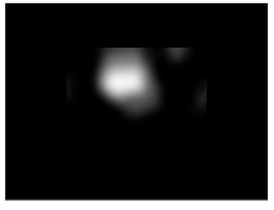
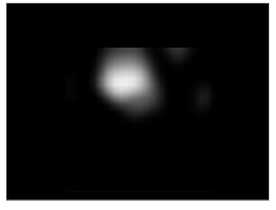
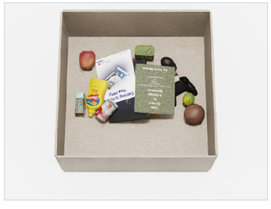
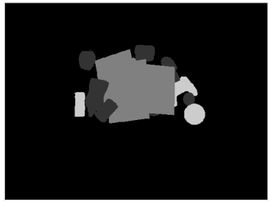
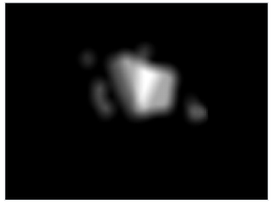
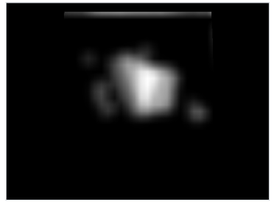
Proposed fusion training

Freeze streams; learn the fusion head.

No fusion result is shown.

Ground truth and predictions on held-out scenes

Fixed examples shared across all three streams

Target	Scene RGB	Similarity		Occlusion		Complexity	
Center reference	Observed image	Ground truth	Prediction	Ground truth	Prediction + workspace	Ground truth	Prediction
 **book_1** center view · scene 01	(a) 						
 **fruit_1** left view · scene 03	(b) 						
 **packaged_food_1** right view · scene 17	(c) 						
 **toy_1** top view · scene 01	(d) 						
 **toy_4** bottom view · scene 07	(e) 						

GT: native image
Prediction: 30 × 40 grid, expanded

